
CSI132 - Discrete Structures II
Tutorial - Primes and GCD
ATTEMPT ALL QUESTIONS
DURATION: 2HRS

1. QUESTION ONE.

- (a) Use the test for primality to determine whether the following numbers are prime or not.
 - (i) 667
 - (ii) 557
 - (iii) 9269

- (b) Find gcd and lcm of the following pairs.
 - (i) $3^7 \cdot 5^3 \cdot 7^3, 2^{11} \cdot 3^5 \cdot 5^9$
 - (ii) $2^2 \cdot 3^3 \cdot 5^5, 2^5 \cdot 3^3 \cdot 5^2$
 - (iii) $2^2 \cdot 7, 5^3 \cdot 13$

- (c) Determine whether the integers in each of these sets are pairwise relatively prime.
 - (i) 21, 34, 55
 - (ii) 14, 15, 21

- (d) Find $\gcd(1000, 625)$ and $\text{lcm}(1000, 625)$ and verify that $\gcd(1000, 625) \cdot \text{lcm}(1000, 625) = 1000 \cdot 625$.

2. QUESTION TWO.

- (a) Use Euclidean algorithm to compute the greatest common divisors listed below.
 - (i) $\gcd(111, 201)$
 - (ii) $\gcd(252, 356)$
 - (iii) $\gcd(1000, 5040)$

- (b) Given the values of p and q below:
 $p = 2^5 3^4 11^7 13^{10}$, and $q = 3^6 11^5 13^{12} 17^4$.
 - (i) What is their greatest common divisor $\gcd(p, q)$?
 - (ii) What is their least common multiple of $\text{lcm}(p, q)$?
 - (iii) Verify the $\gcd(p, q) \cdot \text{lcm}(p, q) = pq$.

- (c) If the product of two integers is $2^7 3^8 5^2 7^{11}$ and their greatest common divisor is $2^3 3^4 5$, what is their least common multiple?

- (d) Show that the following statement is false:
There is a positive integer n such that $n^2 + 3n + 2$ is prime.